Repeated games

Will it end?

Play
Bid High or Low. Then roll the D12.

→

Shorten
Carry on only on a 9 or above.

→

Explain
Why the future changes today.

Bribentives for the podium



	High	Low
High	3	0
Low	5	1

Your payoff as the row bidder.

Played once, what should you do?

Played once, what do you both get?

And if you had both bid High?

Play it again, and again.

Roll the D12 after every round.

On a 1 or a 2 the game ends. Otherwise play another round.

Keep a running total. Nobody knows how many rounds there will be.

The game ends on a 1 or a 2, so it carries on with probability

$$\delta = \square$$

The expected number of rounds is $\frac{1}{1-\delta} = \square$

New rule: we carry on *only* on a 9 or above.

$\delta =$ ____ $\qquad \frac{1}{1-\delta} =$ ____

Before we replay: will you bid High as often? ____

Same payoffs, same opponent, shorter future.

Grudger plays High until you play Low, then Low for ever.

You conform, and play High for ever

$3 + 3\delta + 3\delta^2 + \cdots =$

You deviate to Low in round one

$5 + \delta + \delta^2 + \cdots =$

$$\frac{3}{1-\delta} \geq 5 + \frac{\delta}{1-\delta} \iff \delta \geq \square$$

Our first die gives $\delta =$ $\square$

The second gives $\delta =$ $\square$

In a round you know is the last, the future is worth [ ], so you bid [ ].

Both of you know that, so keep working backwards and cooperation _____.

The die was doing real work: not by making the game long, but by making the end _____.

What we just did in the room is **Question 1** on the Repeated Games page.

(a)	Interpret $\delta = 5/6$ and give the expected number of rounds.	[2]
(b)	Discounted payoffs against Grudger, conforming and deviating.	[6]
(c)	Is cooperation sustained at $\delta = 5/6$?	[4]
(d)	The smallest δ that sustains it, and what our die meant.	[5]
(e)	The same for a forgiving trigger that punishes for two rounds only.	[7]

vknight.org/gt/topics/repeated-games.html

Every stage of today, and the part of the notes it was reading.

What we did	Notes	In the textbook
Bidding High or Low once	**2.2.3**	Games: Definition: Normal Form Game
Playing it again, and again	**7.2.1**	Repeated Games: Definition: repeated game
The die as δ	**7.2.5**	Repeated Games: Infinitely Repeated Game with Discounting
Running totals per round	**7.2.6**	Repeated Games: Definition: Average utility
Grudger, and deviating from it	**7.2.2**	Repeated Games: Definition: Strategies in a repeated game
The threshold on δ	**7.2.7**	Repeated Games: Definition of individually rational payoffs
Examinable	**7.2.3, 7.2.8**	Repeated Games: Subgame perfection of sequence of stage Nash profiles, and the Folk Theorem, both with proof

vknight.org/gtb/chapters/games, vknight.org/gtb/chapters/repeated-games

Quick quiz before you go: vknight.org/gt/topics/repeated-games.html

Before the next class: read the Repeated Games chapter as far as the Folk Theorem, and attempt Question 1.